

Article

Leveraging AI for Network Threat Detection—A Conceptual Overview

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Abstract: Network forensics is commonly used to identify and analyse evidence of any illegal or unauthorised activity in a given network. The collected information can be used for preventive measures against potential cyber attacks and serve as evidence acceptable in legal proceedings. Several conventional tools and techniques are available to identify and collect such pieces of evidence; however, most of them require expensive commercial resources, longer investigation times, and costly human expertise. Due to modern networks' diverse and heterogeneous nature, forensic operations through conventional means become a cumbersome and challenging process. This calls for a new look at how network forensics is approached, considering contemporary approaches to network analysis. In this work, we explore artificial intelligence (AI) techniques based on contemporary machine learning (ML) algorithms such as deep learning (DL) to conduct network forensics. We also propose an investigation model based on AI/ML techniques that can analyse network traffic and behavioural patterns to identify any prior or potential cyber attacks. The proposed AI-based network forensics model speeds up the investigation process, boosting network monitoring without human intervention. This also aims to provide timely and accurate information to network administrators for quick and effective decisions, enabling them to avoid and circumvent future cyber attacks.



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Keywords: digital forensics; network forensics; artificial intelligence; machine learning; deep learning; internet of things

1. Introduction

1.1. Background and Motivation

In an era marked by unprecedented technological advancements, the cybersecurity landscape [1] is continuously evolving, presenting both new challenges and opportunities. As connectivity and network technologies advance, adversaries are increasingly exploiting gaps and vulnerabilities, leading to a rise in various cyber attacks [2]. Notably, the frequency and sophistication of distributed denial-of-service (DDoS) and ransomware attacks [3] have surged, highlighting the urgent need for robust security measures across corporate, commercial, and domestic networks. Figure 1 shows a comparative analysis of annual DDoS attacks from 2018 to 2023. The Internet of Things (IoT), creating an extensive global computing network that connects individuals and entities to the Internet [4], further complicates the security operational landscape. While offering significant benefits, this interconnectedness also introduces new vulnerabilities, raising the critical importance of implementing comprehensive cybersecurity strategies to protect against these evolving threats.

The fundamental concern revolves around enabling self-governing interactions and facilitating the secure exchange of valuable information across distinct and uniquely identifiable real-world entities, i.e., applications and devices. These real-time entities span various

industries such as agriculture, transport, healthcare, and manufacturing, with future implications for even everyday items such as food, furniture, and paper-based documents, as anticipated by the US National Intelligence Council [5]. However, with billions of connected devices, the security and privacy of users become vulnerable due to inadequate security measures, making them susceptible to rapid system breaches and mishandling of valuable information [6]. Additionally, unresolved operational issues can further exacerbate security risks and weaken the trustworthiness of the deployed multi-technological paradigm.

These security concerns also extend to other forms of communication, especially in different networks under various forms and attack vectors. For instance, according to the recent State of Security report [7], 86% of respondents indicated that the current geopolitical climate is increasing the likelihood of their organization being targeted. In [8], threat intelligence data show that ransomware cases increased by 73% resulting in ransomware remaining a constant threat (over 1 billion records compromised and still growing in 2024 [9]), receiving significant attention from governments worldwide due to its widespread effects on everyday activities including healthcare (recently, one of the healthcare firms paid \$22 million in ransom [10] to restore patients data), air travel, automated vehicles, and even daily purchases of grocery items. In terms of consumer services, 70% of ransomware attacks result from web traffic and applications. Therefore, experts predict that by 2031, ransomware attacks will occur every 2 s, impacting organizations, consumers, and devices [11].

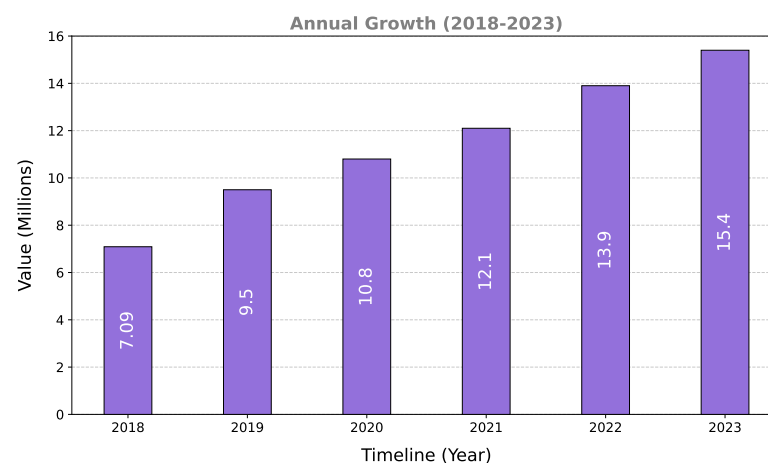


Figure 1. A comparative analysis of annual DDoS attacks from 2018 to 2023 [12].

To combat such catastrophic threats, network forensics (NF) [13,14] is employed to analyse any network traffic when there is a suspicion of cybercrime, such as the proliferation of malicious software designed to steal data. Falling within the purview of digital forensics (DF), NF involves the monitoring and analysing of computer network traffic for data collection. Under the NF/DF (NDF) intelligence approach, incident responders seek technologies or data containing criminal evidence, enabling businesses to enhance their security posture and implement the necessary corrections effectively. Unlike traditional DF, NF integration allows cyber professionals and crime investigators to trace communications and establish a timeline of network events orchestrated through a network-controlled system. Beyond criminal investigations, NDF intelligence is frequently used to analyse network events, trace the origins of ransomware and other security-related incidents, and identify unauthorized network access. By utilizing artificial intelligence (AI) and machine learning (ML), researchers can identify irregularities or potentially harmful behaviours, offering a dynamic and proactive security solution that can adapt and respond to emerging threats.

1.2. Comparative Literature Review

Recent research efforts in the field of NDF have focused on addressing the challenges posed by increasingly sophisticated cyber threats, as evidenced by recent studies [15–20]. While recent scholarly endeavours have incorporated sophisticated techniques such as ML and AI, they face challenges adapting to complex attacks and depend on high-quality training data. Our framework is proposed to address these limitations by employing contemporary AI and ML to augment conventional NDF methodologies, thereby emphasising the imperative for continued research in response to the dynamic nature of cyber threats. In Table 1, we systematically compare our proposed framework with the extant literature, including the works in [21–24] within the NDF domain.

Table 1. NDF comparative table with existing solutions.

Ref.	Recent Contributions	Limitations	Potential Outcomes
[21]	Scope: The authors propose a lightweight, privacy-preserving BERT-based model tailored for IoT/IIoT devices. The model leverages the natural language processing (NLP) capabilities of LLMs like BERT to analyse data traffic, detect anomalies, and classify cyber threats in real-time.	The authors focus on adapting existing NLP-based models for IoT cybersecurity. However, the proposed model may not fully address threat vectors like sophisticated ransomware or AI-driven attacks which require more complex modelling and resources.	Potential enhancements: The authors proposed fine-tuning, continuous updating, real-world training, and autonomous cybersecurity mitigation. Suggested integrations: Integration of federated learning to enhance privacy preservation, design of complex architecture including cross-layer security mechanisms to handle more complex cyber threats in dynamic environments.
[22]	Scope: The authors proposed a comprehensive approach to leverage explainable AI (XAI) in intrusion detection in IoT to enhance cyber defence capabilities. Various methodologies for intrusion detection are examined, focusing on machine learning and deep learning approaches that leverage large datasets for anomaly detection.	The authors identified that the effectiveness of machine learning models relies heavily on the quality and volume of training data, which can be challenging to obtain in diverse IoT environments.	Potential enhancements: The proposed approach enhances the explainability of machine learning models used in intrusion detection systems (IDS) to ensure that stakeholders can comprehend and trust the decisions made by these systems. Future directions: The fast pace of technological advancement in IoT devices and attack methods creates a moving target for security solutions, necessitating ongoing research and adaptation.
[23]	Scope: The authors highlight hybrid machine and deep learning approaches and emphasise how this hybrid model can enhance detection accuracy and reduce false positives in IDS. The authors investigated various algorithms from both domains and their integration, including decision trees, support vector machines, CNNs, and RNNs.	Integrating multiple algorithms can introduce complexity, making model training and maintenance more challenging. Hybrid models may require significant computational resources, which can be a limitation in environments with constrained resources.	Potential enhancements: Further exploration of anomaly detection capabilities in hybrid models could improve their adaptability to novel threats that are not present in the training datasets. Future directions: The fast pace of technological advancement in both IoT devices and attack methods creates a moving target for security solutions, necessitating ongoing research and adaptation.
[24]	Scope: The article discusses how network security devices, such as firewalls, VPNs, and IDS, play a critical role in defending IT environments. These devices act as a “front door” to protect the internal systems from external threats.	Even though these devices are meant to protect networks, they are still prone to vulnerabilities. Another challenge is the inconsistent application of security patches and blind spots in detection, i.e., the existing struggle to detect sophisticated multi-vector attacks. It always remains a limitation that hackers shift focus and attack strategies to find a gap.	Potential enhancements: Enhanced Zero-Trust security models in conjunction with network security devices. Future directions: Integration of AI/ML in these network devices to improve real-time threat detection and automated responses, making them more adaptive to evolving threat landscapes.
Our work	Scope: The scope of this study is to present a practical investigation framework for enterprise security and discuss its conception stage to establish a state-of-the-art model, in order to verify every access request and reduce the risk of outsider/insider threat or lateral movement after an initial breach.	The focus of this work is limited to the following areas: (i) The literature findings related to gaps and challenges in the field of network threat detection. (ii) Identifying contributing factors and use cases associated with the attack surface. (iii) Examining how contemporary AI and ML techniques serve as hybrid models to enhance traditional network detection frameworks. (iv) Defining the boundaries of our potential research scope for this study.	Potential enhancements: The extension of this work is to model stages of CTDF as shown in Figure 6. Future directions: Readers are encouraged to read the proposed model’s expected outcomes in Figure 9.

1.3. Contribution and Organization

We intend to formulate an approach within the scope of our NDF research to integrate AI and deep learning (DL) models for network traffic analysis. The focus is on detecting abnormal traffic patterns, such as irregular packet sizes, encoded data, and encapsulated traffic that may be used to conceal malicious activities. As illustrated in Figure 2, this work is at the conception stage and various specialist AI models for this purpose will be explored in future extensions of this work (some of the considerations are large language models (LLMs) [25] including transformer models like Google BERT, which have the capability for threat detection and analysis. LLMs can analyse network data in real-time to detect anomalies and potential threats and can recognise patterns that are indicative of cyber attacks, such as reconnaissance, lateral movement detection, and unusual network traffic [21]).

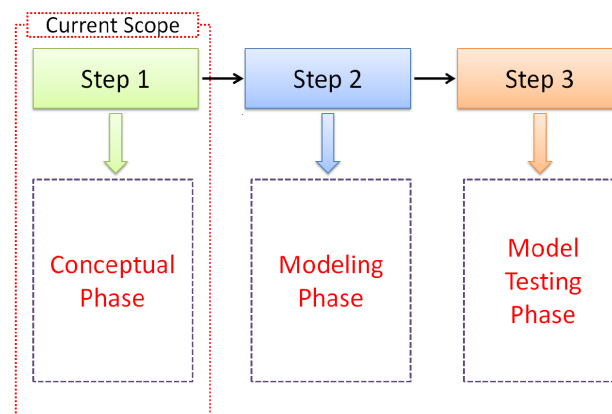


Figure 2. Projected stages of our research work. The first stage focuses on conceptual development and forms the basis of this work.

Some notable contributions of this work are as follows:

- This paper presents a concise overview of the impact of AI and ML on the NDF-intelligence-based forensics approach. The aim is to provide readers with an understanding of the innovative facets of AI-enabled DF techniques used in NF. Such insights hold the promise of influencing and reshaping the landscape of DF in various communication networks.
- Using contemporary ML techniques, we propose a practical investigation framework for enterprise security. The proposed framework aims to automate and analyse network traffic and behaviour patterns for the identification of any prior or potential cyber attacks.
- This paper identifies the main factors and use cases that contribute significantly to the attack surface of a corporate network. Furthermore, some suggestions are provided for addressing these issues through the use of AI/ML techniques. This hinges on identifying gaps in legacy DF systems that could compromise forensic evidence in the proximate future.
- We hope that this paper serves as a starting point for NF professionals and researchers who are looking for the use of AI/ML techniques for NF investigations. The generic framework proposed in this article serves as a template for building a purpose-fit AI/ML model for use in an enterprise setup. Furthermore, the integration of such a system allows the generation of evidence for cybercrime, suitable for presentation in a court of law by professionals.

The rest of this paper is organised as follows. Section 2 details the integration of digital forensics and artificial intelligence, encompassing network forensics, artificial intelligence, and machine learning. Section 3 focuses on research problem formulation, emphasizing the development and application of a common enterprise network security architecture model. Section 4 presents the proposed ML framework. Anticipated future research challenges are outlined in Section 5, addressing emerging complexities and ethical considerations. This paper concludes in Section 6, summarizing key findings, highlighting contributions, and discussing the significance of the integrated approach in the dynamic field of forensics and AI.

2. Forensics and Artificial Intelligence

AI is generally viewed as a facilitating technology, specifically through its significance in most incident response phases. Such mechanisms adeptly process objective data, generating valid potential hypotheses that can be presented as evidence in a court of law. As shown in Figure 3, NF is devoted to examining and scrutinizing events within a network, covering both historical occurrences and those anticipated in the future. Network forensics works in conjunction with AI and ML techniques, relying on them for critical analysis through their use in pivotal tasks such as feature classification, information representation, and pattern recognition. The integration of AI and ML introduces a sophisticated and automated approach, enabling investigators to skilfully classify features, represent information, and recognise patterns, consequently amplifying the efficiency and effectiveness of network forensic processes.

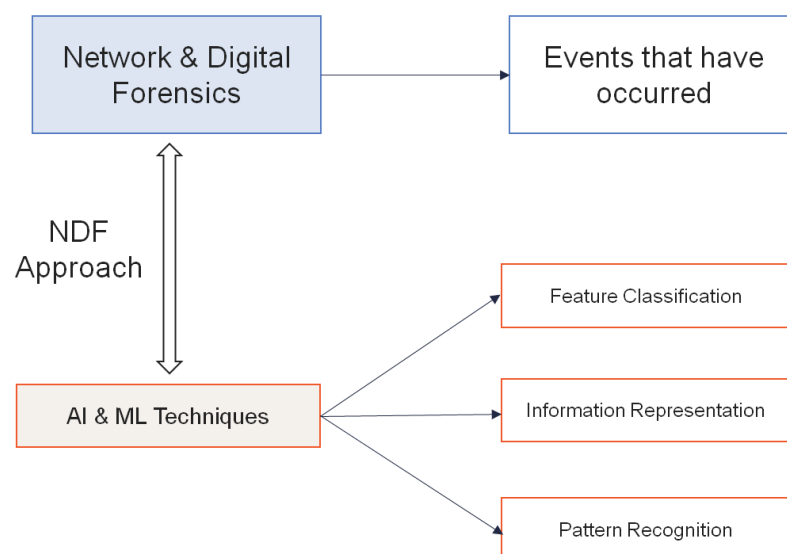


Figure 3. The joint role of network forensics and AI in network security.

2.1. Artificial Intelligence

The concept of AI refers to the capability of machines to perform tasks typically associated with human intelligence, such as natural language understanding, and decision-making. Within the domain of AI, ML, as a specialized branch, employs algorithms that enable systems to improve and learn from data over time. DL, a subset of ML, leverages artificial neural networks to enable computers to learn from extensive datasets, and holds particular significance in enhancing the learning capabilities of machines. AI methods can tackle challenges within the realm of digital forensics, providing a means to address computationally vast or intricate issues within a feasible time frame. The essence here lies in creating computer systems and programs capable of intelligent actions. In simpler terms, it involves developing computer systems or machines that can replicate the behaviour and intelligence of human beings.

In the context of cyber security, AI can help automate threat detection, analysis, and mitigation, reducing the time from detection to resolution. AI systems can also prioritize security alerts and help in identifying root causes to prevent any future issues [26,27]. Furthermore, generative AI can synthesize data from various sources into comprehensive reports that highlight key findings, trends, and potential vulnerabilities [28]. AI techniques can be also harnessed to design and develop intelligent attack detection models for securing diverse systems [29,30], enabling networks to extract insights from unsupervised data and effectively tackle complex problems [31]. However, AI also presents some possible challenges, such as the potential for adversaries to target AI systems [32,33]. Therefore, organizations must ensure that AI-based solutions are regularly updated and monitored to prevent them from becoming exploitable. In another study [34], an IDS model is proposed by researchers, utilizing AI optimization techniques to enhance attack categorization. The model includes automatic hyperparameter selection through Golden Eagle Optimization (GEO) and AI frameworks to improve classification performance.

2.2. Machine Learning

ML stands out as a pivotal technology in the Industry 4.0 or onwards revolution [35–37], enabling systems to improve their performance through experience without the need for explicit programming. In the realm of cybersecurity, ML plays a crucial role in extracting valuable insights from both organized and unstructured data from various sources. These data serve as the foundation for constructing intelligent applications, including the prediction of zero-day attacks, malware detection or phishing, anomaly detection or cyber attack, and intrusion detection. Therefore, incorporating AI and ML into predictive threat intelligence significantly enhances the cybersecurity resilience of different sophisticated networks, particularly corporate/government, enabling a shift from reactive to proactive defence by detecting and mitigating threats early [38]. ML is a diverse field encompassing a wide array of techniques, which can be broadly categorized into traditional and contemporary techniques as described below.

Traditional vs. Contemporary ML Techniques

In the traditional approach of supervised learning [39], the algorithm develops the ability to generalize from labelled datasets to map input data with their corresponding outputs. In contrast, unsupervised learning [40] tackles unlabelled data, with its objective being the relationship and identification of patterns or structures inside the data, all without output-related guidance. Decision trees, a versatile model, are used for both classification and regression tasks, where they make decisions based on a series of questions about the input data. Support vector machines (SVMs) [41] are another supervised learning model utilized for classification and regression analysis that determine a hyperplane optimally separating data points into different classes. Other traditional techniques can include regression analysis [42], K-means clustering [43,44], and principal component analysis [45]. In the subset of contemporary techniques, DL emerges as a distinct subset of ML, characterized by the utilization of neural networks having multiple layers; commonly referred to as deep neural networks. This specialized approach has showcased notable effectiveness, especially in tasks such as image and speech recognition. In the context of cyber security, DL models can be used to analyse vast amounts of data, including data from the network and servers, to build offensive and defensive strategies, such as automated penetration testing, intrusion discovery, and vulnerability and malware detection in real-time. We refer the interested reader to the works of [46–48] for further details.

2.3. Network Forensics

NF plays a crucial role in assisting investigators in extracting valuable evidence from network traffic. Investigators aim to uncover specific information within the network traffic that serves as evidence in their investigations. This includes identifying patterns, anomalies, and key elements within the network stream [49]. NF also enables the collection,

analysis, and alignment of network traffic data, allowing investigators to examine traffic forms between different machines. The sought-after information often includes evidence related to potential security incidents, unauthorized access, data breaches, or other illicit activities occurring within the network environment. Additionally, investigators may use it to uncover insights into the communication and interactions between networked entities, aiding in the reconstruction of events and understanding the nature of potential threats or cybercrimes [14].

2.4. AI in Network Forensics

The rising adoption of AI techniques is revolutionizing the automated detection of attacks and enhancing the accuracy of network traffic classification [50,51]. Network forensic investigations span a wide scope, from Internet-wide analyses to examining the network traffic of individual devices. Networks exhibit diverse characteristics influenced by factors such as applications, equipment, scale, architecture, and so on, posing challenges for ML approaches that require training for each network independently. However, ML-trained models may face accuracy challenges across varying network typologies. Deriving valuable insights and patterns from security data, machine and deep learning models employ complex transfer functions, methods, and rules. This training equips machines to predict threats as well as risks proactively at the early stages of development. Nevertheless, conventional ML algorithms necessitate large sets of training data for specific tasks, triggering concerns about security and privacy. The prevalent challenge of privacy leakage arises when device data must be transmitted to a centralized third-party server, heightening privacy and security risks. In [52], the exploration of AI paradigms within social media forensics is undertaken, emphasizing digital forensics, its various subdomains, and notable advancements in the field. The principal AI methodologies identified in this study are natural language processing (NLP), graph neural networks (GNNs), and generative adversarial networks (GANs). AI adoption in NF accelerates investigations by identifying cybercrime data patterns. Important AI techniques include knowledge representation, expert systems, and pattern recognition. Hence, adaptable methods are crucial to handle evolving data and scalability in legal contexts [53].

Integrating AI into network forensics further improves cyber incident detection and analysis using tools like anomaly detection algorithms and machine learning-based intrusion detection systems. This leads to better threat attribution and faster response times, though challenges remain in handling large data volumes and reducing false positives [54]. The incorporation of artificial intelligence into network forensics, exemplified by tools such as the Trustworthy Cyber Attack Detector (TCAD) tool, augments the fight against cybercrime by enhancing the accuracy of detection and optimizing the efficiency of investigations [55]. This integration facilitates a proactive stance in addressing emerging threats while upholding ethical standards and ensuring privacy.

3. Conceptual System Model

We consider an enterprise network architecture focused on security controls against various cyber threats, as shown in Figure 4. It begins with secure remote access for authorized users, safeguarded by an external layer featuring a next-generation firewall with intrusion prevention systems and an anti-malware engine. Moving inward, a perimeter firewall with anti-malware capabilities connects to three network types, DMZ (demilitarized zone), enterprise network, and corporate users, each surrounded by its firewall for network segmentation.

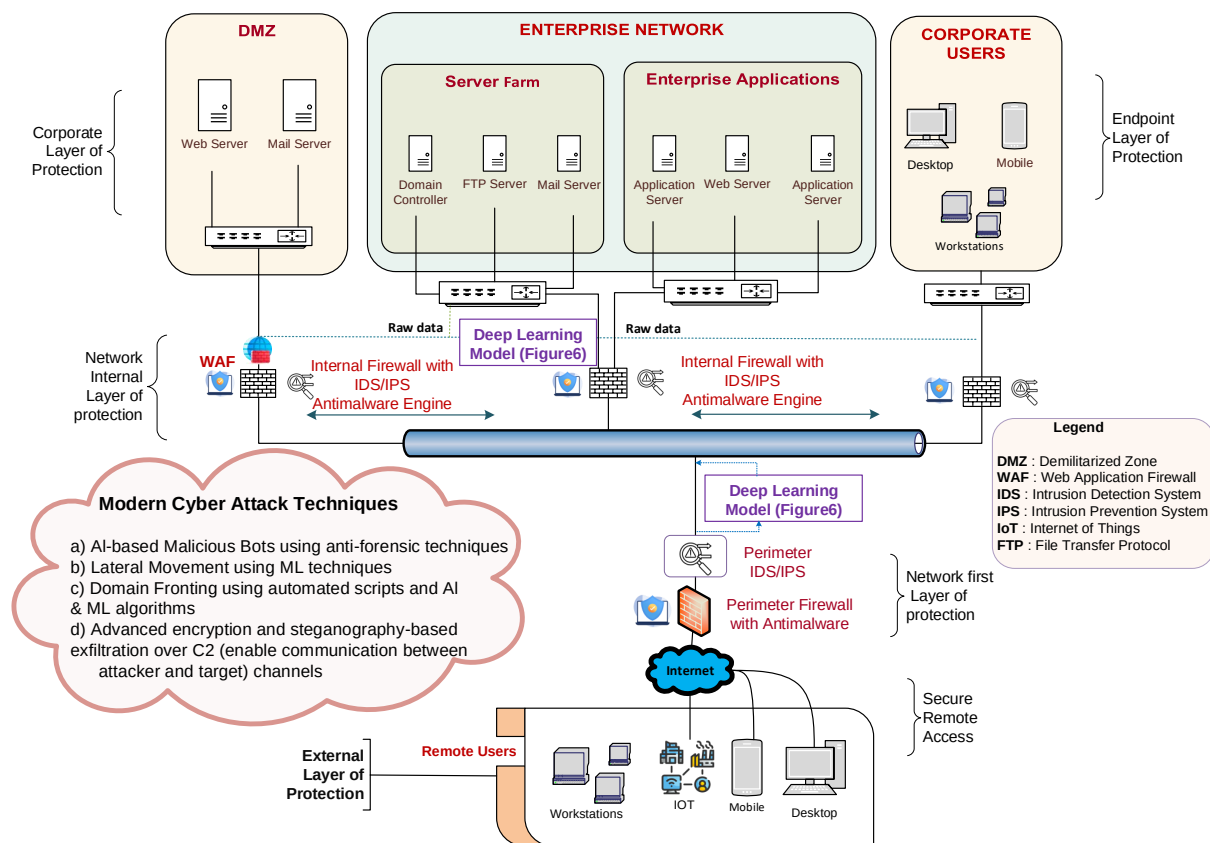


Figure 4. A common enterprise network security architecture model.

- The DMZ acts as a buffer, hosting public-facing servers to segregate them from the internal network.
- The enterprise network includes a server farm for hosting internal enterprise applications.
- The corporate users, internal to the network, are protected by a corporate layer which integrates firewalls, intrusion detection and prevention systems (IDS/IPS), and anti-malware engines.

This multi-layered approach ensures a robust defence against cyber threats, with firewalls, anti-malware, and IDS/IPS operating at different levels. The depicted enterprise network diagram showcases a multi-layered security approach designed specifically for ML adoption. The proposed framework of network forensics using AI is based on two components. The first component is a network that has been affected by a cyber attack, and the investigator needs to know various data collection points at the network where useful data related to the cyber attack can be collected. The second component is the AI-based model that will take the collected data and analyse them to answer the forensic questions of interest. In the following, we first describe several preliminaries encompassing the various factors and use cases that contribute to the Network Attack Surface (NAS) of a corporate network. Next, we explain both components of the proposed framework in light of these factors.

3.1. Preliminaries

In the following, we explain a few of the main terminologies and concepts that are critical in the background of network forensics and are used sparingly in this work in the context of Figure 4:

- *Network Architecture*: The design and structure of the network, including the number and types of devices, their interconnections, and the presence of firewalls among other security measures; will effectively have a high influence on the attack surface.
- *Network Services and Protocols*: The services and protocols running on the network, such as web servers, email servers, and file-sharing protocols. Each service or protocol introduces potential vulnerabilities that could be exploited by attackers.
- *Software and Applications*: The presence of various software and applications on networked devices adds to the attack surface, as outdated or insecure software can be a target for exploitation.
- *User Accounts and Permissions*: The number of user accounts, their permissions, and the strength of authentication mechanisms affect the attack surface. Weak or compromised user accounts can provide unauthorized access to the network.
- *Connected Devices*: The proliferation of connected devices, including IoT, contributes to the attack surface. These devices generally have limited security features and can be exploited to gain access to the network.
- *Remote Access Points*: The presence of remote access points, such as VPN connections or other remote access technologies, expands the attack surface. If not properly secured, these points can be exploited by attackers.
- *External Interfaces and APIs*: Interfaces and Application Programming Interfaces (APIs) that connect the network with external services or third-party applications introduce potential vulnerabilities whose insecurities may be targeted for attacks.
- *Physical Security Measures*: Physical security measures, or the lack thereof, also contribute to the attack surface. Unsecured physical access points (such as plug-and-play devices and other ports including console port) to network infrastructure may lead to unauthorized access.
- *Security Policies and Configurations*: The effectiveness of security policies, configurations, and monitoring mechanisms directly impacts the attack surface, where misconfiguration or weak policies may expose vulnerabilities.

Overall, these terminologies contribute to the conception of our proposed DL model.

3.2. NAS Use Cases

In this subsection, we state various use cases to highlight the diverse tactics that may be employed to target and achieve malicious objectives, ranging from data theft and system compromise to evasion of detection mechanisms. Vigilance and robust security measures are essential to mitigate the risk of such attacks. These use cases showcase attractive target areas that need further investigation in the context of our proposed corporate model and are listed in Table 2. Such use cases serve as a basic example for readers' understanding and also provide a list of perceived challenges for the proposed DL model under discussion. Owing to the evolving cyber attack surface, it is always a challenge for any research investigation to predict potential targets for an adversary.

Table 2. NAS use cases showcase attractive target areas.

Use Cases	Threat Vector
Ransomware Active Attack on the Database	Encrypting sensitive data within a database and demanding ransom for decryption.
OS File Read	Unauthorized reading of critical operating system files for reconnaissance or data theft.
Malware Deployment	Deploying malicious software to compromise system integrity or facilitate further attacks.
Privilege Escalation	Elevating user privilege to gain unauthorized access or perform malicious actions.
Audit Tampering	Manipulating or disabling auditing mechanisms to cover tracks and evade detection.
Command Execution	Running arbitrary commands on a compromised system to execute malicious activities.
Credential Extraction	Stealing login credentials to gain unauthorized access to systems or services.
Data Exfiltration	Illegally transferring sensitive data from a targeted system to a remote location under the attacker's control.
Excessive Record Access	Unauthorized retrieval of large amounts of data from databases or storage systems.
Exploit Active Directory Security	Targeting vulnerabilities in Active Directory to compromise user accounts or escalate privileges.
Excessive Failed Logins	Attempting to gain access to a system or service through brute force or credential stuffing attacks.
Service Account Abuse	Misusing privileged service accounts to perform unauthorized actions in a network.
Non-standard Account Access	Utilizing non-standard or overlooked user accounts to bypass security.
Machine Takeover	Gaining full control of a targeted machine, potentially for use in botnets or other malicious activities.
Silent Actor	Conducting activities with minimal noise or detection to avoid raising suspicion.
Slow Rate File Access	Deliberately accessing files at a slow pace to evade detection by security systems.
Suspicious File Access	Accessing files or directories in a manner that triggers alerts due to unusual behaviour.
Sensitive Tables Scan	Scanning databases for sensitive tables containing valuable or confidential information.
Dynamic SQL Activity	Utilizing dynamic SQL queries to manipulate or extract data from databases.
Malafide APIs	Exploiting vulnerabilities in APIs to perform unauthorized actions or extract data.

3.3. Generative AI and NAS

GenAI undoubtedly offers numerous benefits and advancements in various fields [56–59]. However, like any technology, it also presents certain security risks, especially when it comes to its application in network security. Here are some potential GenAI security risks:

- *Adversarial Attacks:* GenAI models can contribute to adversarial attacks, where attackers manipulate input data to deceive the implemented AI model into making incorrect predictions or classifications [60]. In the context of network security, this could be exploited to bypass intrusion detection systems or fool AI-based network defences.
- *Data Poisoning:* AI models rely heavily on training data to learn and make decisions. If the training data are compromised or poisoned with malicious information through GenAI models [61], it can lead to biased or inaccurate AI-driven security decisions, potentially weakening network defences.

- *Model Inversion Attacks*: Attackers can attempt to reverse-engineer AI models with the help of GenAI by exploiting vulnerabilities in the implemented model's architecture or by probing the model with crafted inputs [62]. This could reveal sensitive information about the model's inner workings, which can then be exploited to undermine its effectiveness in detecting threats.
- *Privacy Concerns*: AI models may inadvertently leak sensitive information in the data they process. In the context of network security, this could include revealing patterns of network traffic or user behaviour that could be exploited by attackers with the help of GenAI to launch targeted attacks or breaches [59].

To mitigate these GenAI security risks, and therefore avoid exploitation of the NAS, our proposed NDF model seeks the organizations to implement the related assessments and practices to create a supportive environment for the overall architecture.

3.4. Network Model

Let us reconsider our enterprise network model shown in Figure 4, which has heterogeneous users with various connections in the network. Several layers of security are implemented in any given network. For example, we have several protection layers protecting servers, enterprise applications, corporate users, and other network nodes. The end users are able to connect to various nodes as per their access levels. The common entrance point of an intruder is through the Internet, by impersonating as either a legitimate user or using the stolen credentials of a legitimate user. When a cyber incident occurs in such a network, the intruder tries to leave no fingerprints of its activities in the network. However, a network forensic investigator can collect data at various network points and give the task of initial assessment to the AI model to identify any anomalies in the data. We describe such an NDF model in the following sub-section, as such a model can play a pivotal role in the subsequent analysis of the attack.

Although next-generation firewalls (NGFWs) [63] and deep packet inspection (DPI) [64,65] are indeed powerful tools for network security, AI serves a different purpose, focusing on enhancing the ability to understand and process huge data at wire-speed. Some of the reasons why AI is being explored for network traffic analysis which would have strong use cases are listed in Table 3. Thus, while NGFWs with deep packet inspection (DPI) are crucial for network security, LLMs complement these tools by providing advanced language-based analysis, behavioural insights, and enhanced threat intelligence capabilities that are beyond the scope of traditional firewalls [66].

Table 3. Overview of LLM capabilities in cybersecurity.

Key Areas	Description
Contextual Understanding and Analysis	NGFWs are designed to identify known threats based on patterns and signatures but may struggle with detecting sophisticated attacks requiring contextual understanding. LLMs can analyse vast amounts of text-based data (e.g., emails, logs, social media) to detect phishing attempts or social engineering.
Threat Intelligence and Communication	AI can assist in processing and analysing threat intelligence data, summarizing reports, and generating actionable insights to identify malicious or suspicious activities that have not been seen before in the network or traffic patterns.
Behavioural Analysis and Anomaly Detection	LLMs can analyse user behaviour or communication patterns to detect anomalies, indicating possible security breaches or insider threats. This capability surpasses NGFWs, as LLMs adapt to the nuances of human behaviour and language.
Addressing Emerging Threats	LLMs, with their ability to process and understand unstructured data, help in identifying and predicting new attack vectors that NGFWs may not yet be able to detect due to a lack of signatures or patterns.

3.5. NDF Model

The authors in [67] presented an analysis of challenges within network forensics using machine learning (ML) and deep learning (DL) models. They identified prolonged processing times as a significant issue associated with multi-layer DL models. To address this, there

is a need for lightweight algorithms characterized by low computing costs, and capable of handling multifaceted problems efficiently. Additionally, the growing prevalence of data encryption and protocol tunnelling introduces modern challenges for security specialists. Another critical area that remains understudied is fault management, as much of the existing literature focuses on deep learning applications beyond forecasting and traffic flow categorization. This gap highlights the necessity for further research and development in fault management using DL techniques within network forensics. In [68], the authors introduce an intrusion detection system (IDS) that utilises ML and DL paradigms. This innovative framework leverages various recurrent neural networks (RNNs), including simple RNNs, gated recurrent units (GRUs), and long short-term memory (LSTM). The distinct advantage of employing RNN-based classifiers lies in their ability to temporally memorize past states and effectively apply them to ongoing computations. The authors in [69] employ a diverse set of ML algorithms, including extreme gradient boosting (XGBoost), gradient boosting, convolutional neural networks (CNNs), support vector machine classifiers, naive Bayes classifiers, and decision tree classifiers. Through training multiple models, the researchers identify the best fit for a particular dataset, enabling the detection of various malware strains and vulnerabilities. The study specifically captures and analyses factors determining malware presence, shedding light on supply, functionality, and capacity outcomes associated with different malware specimens.

Figure 5, taken from [70] as an example, shows a capture of malicious network traffic under malware campaigns. Utilizing search engine optimization (SEO) poisoning techniques, the cyber threat actors engaged victims into downloading malicious software from compromised websites through drive-by compromises. Row 1 of the figure shows the fake website, in which the threat actor engages and redirects the victim to the malicious website that is shown in Row 2. Once the victim clicks on the malicious link on the website in Row 2, it downloads the malicious file and installs it on the victim's workstation, which establishes the connectivity with cyber threat actors' command-and-control (C2C) communication infrastructure. The C2C server, hosted on the Internet, executes post-compromise tasks such as downloading additional malicious software, stealing data, or establishing remote control. After establishing the connectivity, the malware tests its connectivity with C2C traffic for keep-alive purposes as shown in Rows 3–24. The above is an example of an identified base log that can not only be exploited by adversaries but might also result in extended analysis time for cyber analysts. Our future model, based on the motivation of the current work, will ingest such patterns and their deviations to make optimised decisions as well as incorporate automated threat identification, providing intelligent approaches.

	Time	Source	Destination	Destination Port	Protocol	Server Name	Info
1	2024-01-19 17:58:30	10.1.19.101	35.202.40.193	443	TLSv1.2	trustadvisorygroup.com	Client Hello
2	2024-01-19 17:58:43	10.1.19.101	34.254.14.186	443	TLSv1.2	www.info-wp.konsport.com	Client Hello
3	2024-01-19 18:14:54	10.1.19.101	104.21.3.234	443	TLSv1.2	pagalsongs.in	Client Hello
4	2024-01-19 18:15:14	10.1.19.101	151.101.66.159	443	TLSv1.2	aeesuisse.ch	Client Hello
5	2024-01-19 18:15:35	10.1.19.101	151.101.66.159	443	TLSv1.2	aeesuisse.ch	Client Hello
6	2024-01-19 18:15:57	10.1.19.101	172.67.134.194	443	TLSv1.2	carapedi.id	Client Hello
7	2024-01-19 18:16:18	10.1.19.101	134.209.23.229	443	TLSv1.2	koodocreative.co.uk	Client Hello
8	2024-01-19 18:16:39	10.1.19.101	151.101.66.159	443	TLSv1.2	aeesuisse.ch	Client Hello
9	2024-01-19 18:17:00	10.1.19.101	194.63.249.219	443	TLSv1.2	kragerosiden.com	Client Hello
10	2024-01-19 18:17:21	10.1.19.101	194.63.249.219	443	TLSv1.2	kragerosiden.com	Client Hello
11	2024-01-19 18:17:42	10.1.19.101	104.21.6.113	443	TLSv1.2	carapedi.id	Client Hello
12	2024-01-19 18:18:04	10.1.19.101	103.221.220.28	443	TLSv1.2	brgresidence.vn	Client Hello
13	2024-01-19 18:18:25	10.1.19.101	134.209.23.229	443	TLSv1.2	koodocreative.co.uk	Client Hello
14	2024-01-19 18:18:47	10.1.19.101	87.236.16.91	443	TLSv1.2	psycursions.ru	Client Hello
15	2024-01-19 18:19:11	10.1.19.101	104.21.2.23	443	TLSv1.2	vtorbaza.com	Client Hello
16	2024-01-19 18:19:33	10.1.19.101	151.101.66.159	443	TLSv1.2	aeesuisse.ch	Client Hello
17	2024-01-19 18:19:57	10.1.19.101	194.63.249.219	443	TLSv1.2	kragerosiden.com	Client Hello
18	2024-01-19 18:20:18	10.1.19.101	172.67.131.82	443	TLSv1.2	pagalsongs.in	Client Hello
19	2024-01-19 18:20:38	10.1.19.101	172.67.131.82	443	TLSv1.2	pagalsongs.in	Client Hello
20	2024-01-19 18:20:59	10.1.19.101	104.21.6.113	443	TLSv1.2	carapedi.id	Client Hello
21	2024-01-19 18:21:20	10.1.19.101	104.21.6.113	443	TLSv1.2	carapedi.id	Client Hello
22	2024-01-19 18:21:41	10.1.19.101	87.233.83.78	443	TLSv1.2	poodledoodle-vachtspec	Client Hello
23	2024-01-19 18:22:02	10.1.19.101	87.233.83.78	443	TLSv1.2	poodledoodle-vachtspec	Client Hello
24	2024-01-19 18:22:23	10.1.19.101	87.236.16.91	443	TLSv1.2	psycursions.ru	Client Hello

Figure 5. An example of a malicious network log from ingested data [70].

4. Proposed Machine Learning Framework

In this section, we propose our NDF model exploiting AI/DL-based architecture to analyse the collected data from a network that has been the target of malicious activities. The proposed AI model is envisioned to possess embedded capabilities such as log and traffic analysis, anomaly detection, forensic investigation support and predictive analysis. The ultimate goal is to help analysts identify and analyse the features that shorten their investigation span and the model can be deployed in state-of-the-art environments including cloud, on-premises, and OT. The proposed cyber threat detection framework (CTDF) is shown in Figure 6, and it aims to identify any abnormalities such as anomalies and threat behaviour in the network traffic using machine learning techniques. It is important to highlight here that most of the anomalies and threat behaviours are identified on the basis of different parameters and data information, e.g., source IP, destination IP, server IP, MTU, and frame size. The proposed framework can be incorporated into any existing network, and as such, an example of this has been shown in Figure 4. In the following, we first detail the various inputs derived from the network traffic and then explain the utilization of ML in our framework.

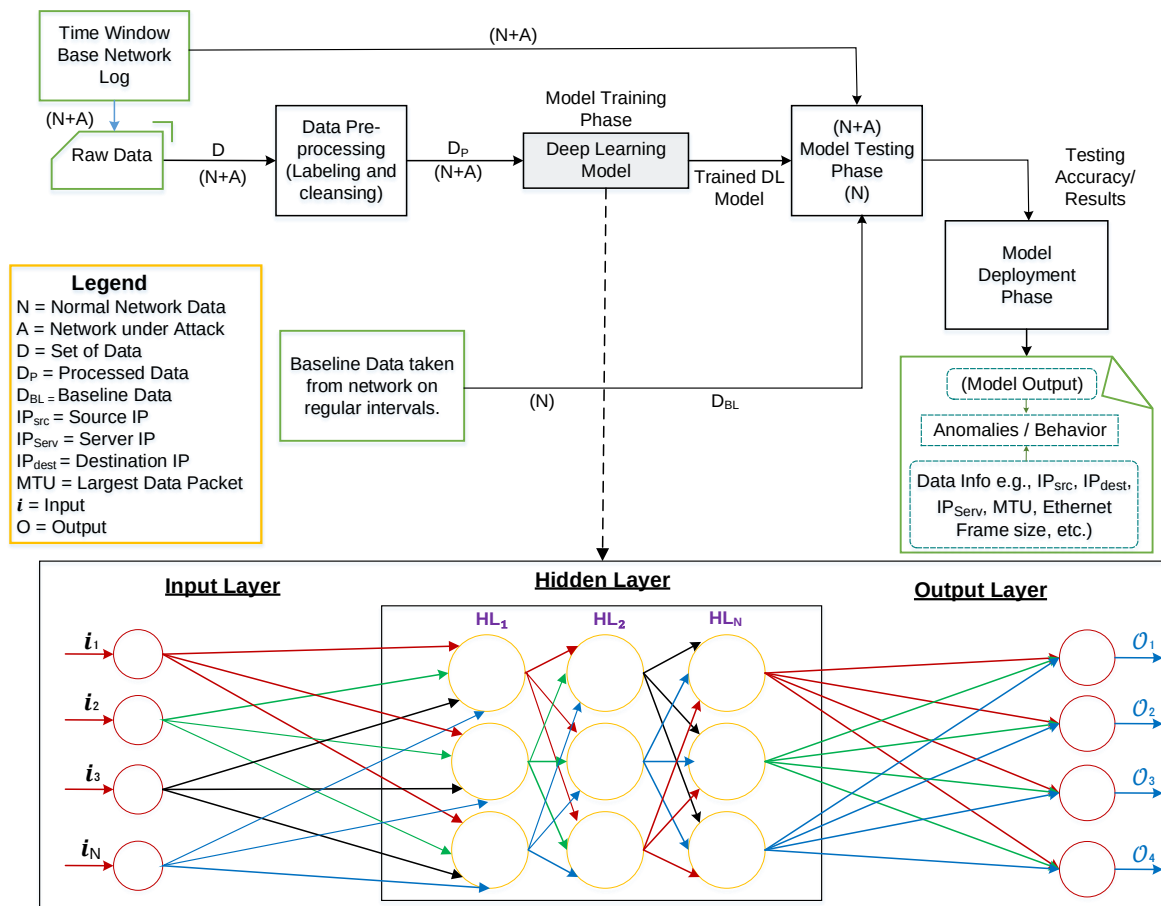


Figure 6. Cyber threat detection framework (CTDF).

The network traffic flow occurrence time is shown in Figure 7. This figure is divided into time slots represented by t , where t_i shows the reference time, t_{i-1} represents the time before the reference time, and t_{i+1} represents the time after the reference time. When the network is operating under normal conditions (i.e., no cyber attack), the network traffic is represented by N . Otherwise, the network traffic under cyber attack in this situation is represented by A . The combination of normal and under-attack traffic is used as input to the proposed CTDF as shown in Figure 6. The network log shows the continuously

flowing traffic ($N + A$) on our network under all scenarios whether the network is operating normally or is considered under attack. The same network traffic is used in the raw data block. The input from the raw data feed is only used when the model needs to be trained. This training can be triggered by either an attack on the network, a change in our network configuration causing changes in network statistics, or it could be scheduled at regular intervals so that the deployed model stays current on the network parameters and statistics. Referring to Figure 7, the baseline data represent the normal network traffic flow (N) during the last t intervals, where t depends on a pre-defined time duration and is assumed fixed in our system. These baseline data (N) establish a working standard, representing the ideal working condition for the network.

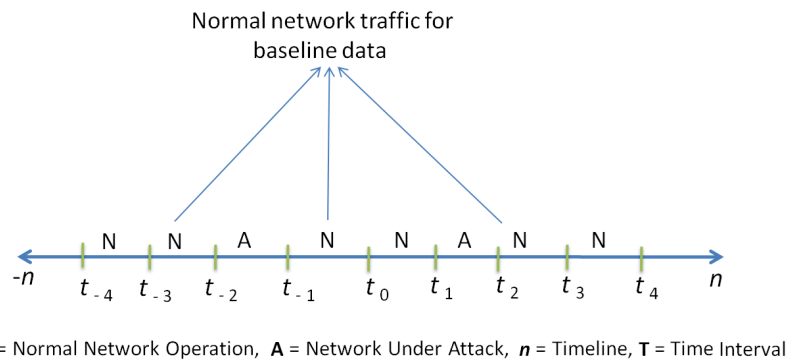


Figure 7. Network dataflow timeline with normal and under attack traffic.

In Figure 6, the collection of raw data is represented as the set $\mathcal{D}$, as shown below

$$\mathcal{D} = \{d_1, d_2, \dots, d_L\}, \quad (1)$$

where L represents the length of the dataset. This data set $\mathcal{D}$ undergoes structuring and labelling in the initial stage, customary in ML operations, to prepare it for analysis. Thus, $\mathcal{D}$ is processed to remove noise, handle missing values, and label instances based on their characteristics. Pre-processing is also used to reduce the network traffic data to certain parameters that are used to train the model, since the network traffic may have various parameters, where we choose only those which are of interest to us for model training. The resultant processed data $\mathcal{D}_p$ can be written as

$$\mathcal{D}_p = \mathcal{D} \setminus \mathcal{R}, \quad (2)$$

where $\mathcal{R}$ is the set of data that has been removed from the collected dataset. $\mathcal{D}_p$ is utilised in training the DL model, which learns to differentiate between normal network behaviour and suspicious activities indicative of potential attacks. This is achieved by learning patterns and relationships that distinguish normal network behaviour from anomalous or malicious behaviour, based on the features extracted from the input data. Subsequently, the trained model undergoes rigorous testing and evaluation to assess its accuracy and effectiveness in identifying anomalies. Testing involves using a separate dataset (e.g., validation or test set) to evaluate the model's ability to generalize to unseen data and accurately identify anomalies. Metrics such as precision, recall, F1-score, and accuracy are calculated to quantify the model's performance.

To validate the initial development of our proposed model, we utilized the NSL-KDD dataset. This dataset comprises 125,971 records weighing 19 MB in its KDDTrain+ set and 22,542 logs weighing 4 MB in its KDDTest+ batch and contains raw network logs, traffic patterns, user activities, and system events. During the data preprocessing phase, the logs underwent cleansing and labelling to prepare the input data ($\mathcal{D}_p$) for our DL model, namely CTDF, as shown in Figure 6. Our initial experiments applied the proposed model to anomaly detection, targeting various network attack types such as Denial of Service (DoS),

port scanning, and data exfiltration attempts. The model yielded the following preliminary performance metrics: an Accuracy of 89.6%, a precision of 88.2%, a recall of 87.4%, and an F1-Score of 87.8%, as illustrated in Figure 8. Although we recognize that these findings are preliminary, these results demonstrate a strong initial performance in detecting anomalous activities and potential threats within the network. The model exhibited robustness in differentiating between normal and malicious behaviors, effectively identifying adversarial actions in the dataset. This early success suggests promising potential for real-world deployment. Further testing utilizing ML algorithms and approaches will be conducted in subsequent iterations, as listed in Section 5.2, to validate and enhance the model's performance. Nonetheless, these initial results provide a solid foundation, supporting the credibility and feasibility of the proposed framework.

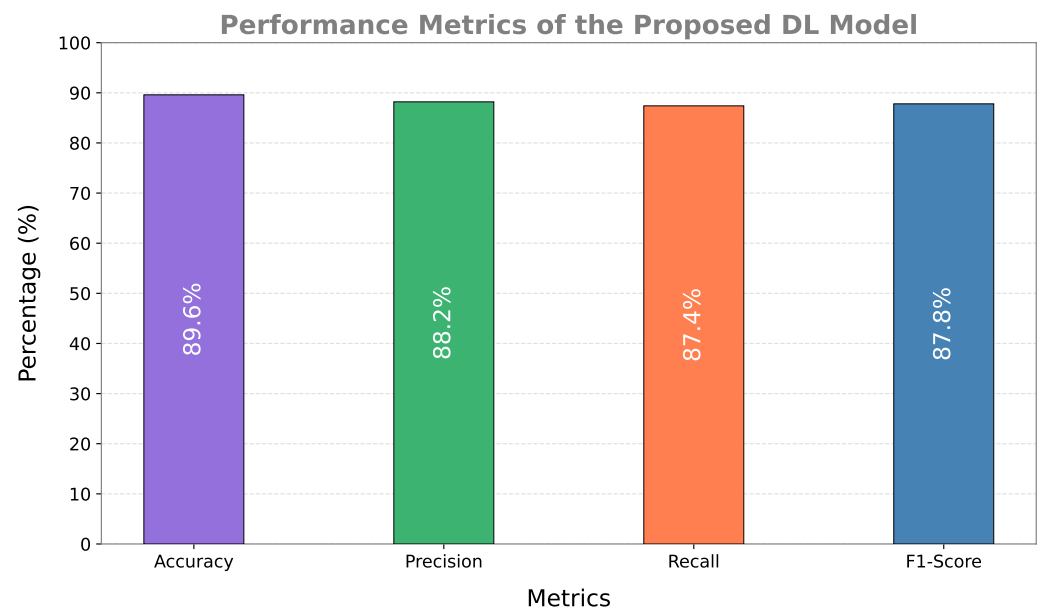


Figure 8. Preliminary performance outcomes of CTDF.

Upon successful testing, the model is deployed within the network environment, where it continuously monitors network traffic in real-time. Detected anomalies and potential threats are reported to network and automation cyber specialists for further investigation, mitigation, and mapping with different globally recognised cyber threat frameworks (CSFs) (MITRE ATT&CK [71–73] and NIST [73,74], among others). The output of the implemented DL model provides work insights into anomalies, unusual behaviours, and potential security risks within the network. Resultantly, this empowers security teams to take proactive measures to safeguard against cyber threats using data information parameters shown as one of the examples of threat anomalies and behaviour classified by the DL model in its output (Figure 6). Continuous monitoring and refinement of the DL model ensure its adaptability to evolving threats, thereby bolstering the overall security posture of the enterprise network. Overall, the model is proposed to be deployed to increase threat detection efficiency. Therefore, Figure 9 is set as the expected outcomes or research direction of the proposed model once deployed in a network security scenario.

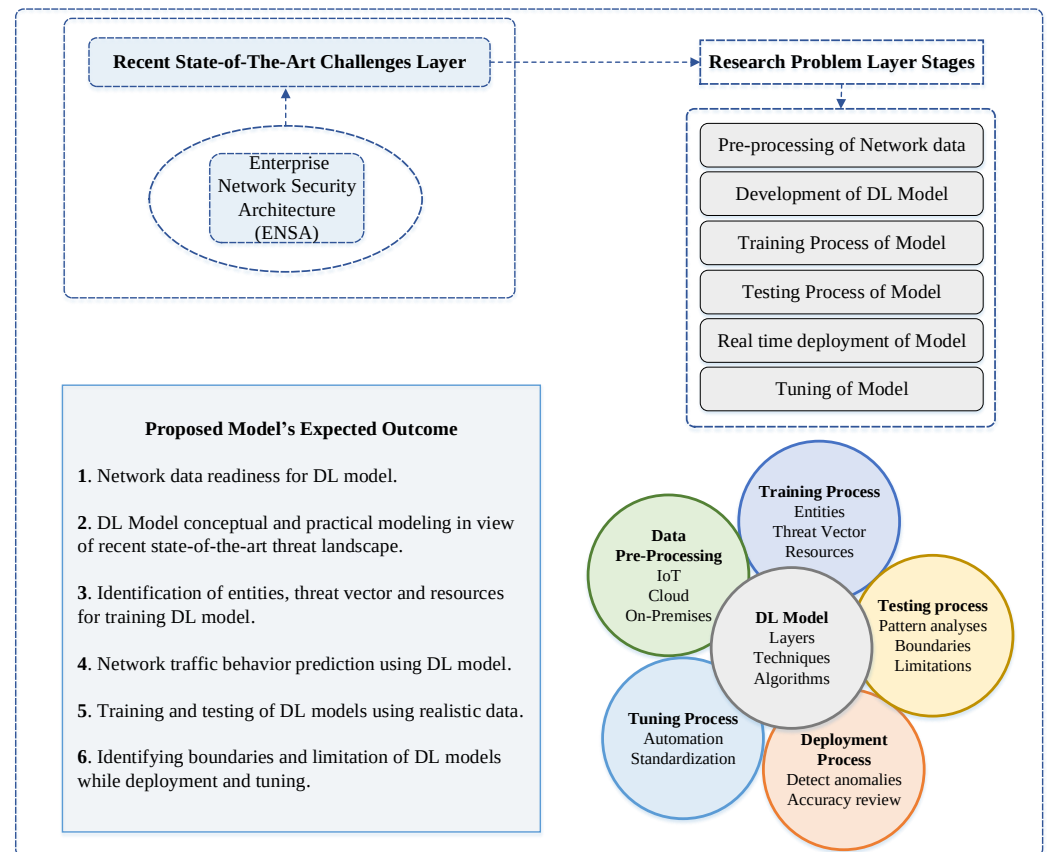


Figure 9. Proposed future research directions and challenges.

5. Discussion and Future Research

The recent surge in DL's adoption is attributed to its remarkable abstraction and generalization capacities, making it a pervasive trend across various domains. In the present era, DL algorithms are extremely useful in various domains like audio processing, autonomous systems, cybersecurity, image recognition, and natural language processing [75]. A compelling rationale for exclusive DL consideration lies in its consistent out-performance over classical ML models. In the context of cybersecurity, DL's capacity to extract intricate patterns from network behaviour provides invaluable insights for robust attack detection [76]. DL also excels in handling the increasing complexity of network traffic amid a growing user base. As network traffic becomes more intricate, DL's proficiency in learning complex data representations becomes pivotal, simplifying the analysis of network traffic complexity autonomously [77]. The thematic summary of our work, as shown in Table 4, presents our initial approach and the *first phase* of conceptualizing the utilization of DL for network forensics. We have identified the distinct advantages of this approach and advocated for its use in modern networks. We have presented various NAS use cases and mentioned some potential security risks where the use of DL in network forensics can immensely benefit the modern-day cyber specialist. We have also presented a theoretical framework for the advocated approach, while the underlying mathematical models and resulting improvement in threat detection is currently under investigation and will be presented in our future works. We also note that similar approaches have been presented in the recent past, including the works in [78–81]. We also refer the interested reader to the works in [47,82] for a detailed overview of recent developments.

Table 4. Discussion and summary of our work.

Thematic Element	Discussion Summary
Introduction	The discussion begins by addressing the rapidly evolving cybersecurity landscape, where technological advancements have led to increased sophistication in cyber threats. This section highlights the urgency for robust security measures to protect critical infrastructure and data. As a motivation, we elaborated on the complexities introduced by the adversaries, which have transformed how organizations operate. This paper stresses the need for comprehensive cybersecurity strategies that encompass not only technological solutions but also considerations for security and privacy implications across various industries, given the pervasive nature of connected devices. The literature review highlights significant recent advancements in NF and cyber threat detection, emphasizing how AI has emerged as a crucial player in enhancing security measures. This section provides insights into how existing research has laid the groundwork for integrating AI into cybersecurity practices, showcasing the promising outcomes of previous studies. Our work establishes the goals and objectives for this study, including defining the research problem and outlining a future path toward developing an advanced model utilizing state-of-the-art techniques.
Forensics and Artificial Intelligence	The discussion points out the vital role of AI and ML in incident response phases of network forensics, automating critical tasks like feature classification and pattern recognition. It emphasizes the importance of these technologies in addressing the challenges posed by complex data environments, ultimately leading to faster and more effective threat detection and analysis.
Conceptual System Model	This section outlines a well-structured enterprise network architecture designed to defend against various cyber threats. By integrating multiple layers of security including firewalls, intrusion detection/prevention systems (IDS/IPS), and anti-malware measures, the architecture proposes resilience against potential attacks. The concept of network segmentation is highlighted as a crucial strategy for minimizing vulnerabilities within the network.
NDF Model and Proposed Framework	The proposed NDF framework is presented as a sophisticated solution for analysing network data to detect malicious activities using AI/DL techniques. This framework aims to support analysts by streamlining the investigation process and providing tools for traffic analysis, anomaly detection, and forensic investigation. The methodology involves pre-processing network data, training deep learning models on critical features, and deploying them for real-time monitoring, ensuring security teams can proactively address cyber threats and adapt to emerging challenges.
Future Research Directions	Defining the boundaries and limitations across the various stages of a proposed layer model is crucial, particularly for each layer from network data pre-processing to model tuning. Each stage demands a comprehensive analysis, investigation, and customized modelling to address specific challenges effectively within the ENSA framework as illustrated in Figure 9.

5.1. Future Research Directions

In this subsection, we present the key components and processes involved in investigating enterprise network security architecture (ENSA) with a focus on utilizing deep learning (DL) models, as shown in Figure 9. These components essentially form the integral pathways that will drive future research in optimizing DL approaches for cyber security and network forensics. Here, we present our view on the future research directions and challenges that come with utilizing AI/ML in network forensics, such that the inherent benefits of these techniques can be fully reaped. We note that investigating the security of ENSA poses certain challenges, specifically in applying network forensics, most of which are different from traditional challenges due to modern-day networks. These challenges are represented pictorially in multiple layers in our proposed future directions as “Research Problem Layer Stages” and shown in Figure 9, where each of them generates a separate objective in securing the proposed architecture. In the following, these research stages are explained in further detail.

5.2. Research Purview

Investigating ENSA in the context of modern challenges due to recent AI and related technologies constitutes the main context for the problem at hand. This can be broken down into various sub-components, as projected in Figure 9, and are further detailed below.

5.2.1. Scope Conception

The scope conception stage considers work boundaries by incorporating the following factors:

- Setting up the work measures, identifying gaps in the literature.
- Conception and utilization of DL models with different phases and stages including pre-processing of network data (e.g., network logs), intelligent decision-making, categorization, segregation, and data mining.

- Data fed from IoT/OT, cloud, and on-premises existing or legacy systems.

5.2.2. Development Process

This includes the development of DL models addressing various challenges considering security layers, techniques, and algorithms including lightweight IoT/OT technology vulnerabilities. We note that this will also involve addressing the protection of industrial networks [83,84] and critical infrastructures [85] to ensure safe and continuous operation.

5.2.3. Training and Testing Processes

In line with the traditional utilization of AI/ML in any domain, continuous training and testing of the deployed models is an essential step [86]. These steps can vary depending upon the application scenario, but will generally consist of a combination of the following steps:

- Identification of entities (users, devices, applications, systems, environment, and other interacting models like GenAI).
- Identification of threat vectors (known threats like malware, phishing, unknown threats, vulnerabilities, and zero-day threats).
- Training involves utilizing available entities, threat vectors, and resources.
- Testing focuses on observing patterns in network traffic, model behaviour, boundary adherence, detection of existing and new threats, and performance optimization [87].

5.2.4. Deployment Process

The critical step ensuring system performance and future feasibility is detecting anomalies and reviewing model accuracy through deployment. In this phase, there can be issues relating to data collection, regulatory compliance, model selection and quality, and scalability. Once these issues have been overcome, a successful model deployment [88] ensures that the DL model is operational and effectively monitoring the network for potential threats.

5.2.5. Tuning

Once the developed model has been deployed, fine-tuning various parameters is required for optimal performance such as the following:

- Adjusting parameters to lower false positives and reduce noise.
- Threshold improvement based on findings from the deployment phase.
- Fine-tuning to ensure optimal performance and intelligent threat detection.

Determining the best parameters might take a long time, especially when the objective function is hard to determine, or the number of parameters to be tuned is large [89]. Some of the most effective techniques for tuning ML model parameters can include grid search, random forest, Bayesian optimization, etc.

Overall, investigating ENSA using DL models involves understanding the challenges posed by modern technologies, developing models to address these challenges, training and testing with relevant data and threat vectors, deploying the models effectively, and continuously tuning them for improved performance and accuracy.

6. Conclusions

As technology advances, the complexities facing digital forensics and cloud forensics have escalated. Data acquisition, for instance, poses difficulties due to the extensive array of digital devices and cloud services, alongside the vast volumes of data requiring collection and analysis. The challenges within digital and cloud forensics can be substantial and diverse, owing to various facets of data analysis, preservation, and security. These challenges are further exacerbated when forensic approaches are applied in the context of network and cyber security. In this work, we have looked at network security from the NDF point of view, providing a heuristic approach to the cybersecurity of modern-day

communications. After looking at the potential use cases, threats, and attack surfaces from an adversary's point of view, we have proposed a framework to utilize AI/ML techniques in analysing network traffic that has been the target of malicious activities. The proposed approach, although still under development, promises to improve network security. Furthermore, taking advantage of the recent developments in the ML/DL domain, it assures of minimum handling required from security personnel. Furthermore, focusing specifically on enterprise network architecture, we have identified several research issues that need prompt attention and action from researchers in both academia and industry, with the potential to spark future works in these directions.

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